



Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University
Alangilan Campus

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Name: ESPARTINEZ, ELAIZA V.

Section: BSCS-1101

ArcHive++

CRUD PROJECT

I. Description

My Project “ArcHive++” is an interactive student records management system designed to perform essential CRUD (Create, Read, Update, Delete) operations. It provides a streamlined interface for managing student information such as student codes, names, programs, contact details, and addresses. Built with simplicity and efficiency in mind, ArcHive++ caters to educators and administrators who require quick and reliable access to student data.

II. Objective

The goal of “ArcHive++” is to deliver an efficient, reliable, and user-friendly system for managing student records in an educational or administrative setting. The specific objective of the project is:

1. To streamline the process of creating, reading, updating, and deleting student records. By offering an intuitive interface for these core functionalities, the system will reduce the complexity and time spent on manual record-keeping, making data management more efficient.
2. To quickly access and manage student data. The system will also ensure that student information is entered and updated correctly, reducing the risk of errors typically associated with manual data entry, thereby improving data accuracy.

III. Output

```
D:\CSProject\CRUD.exe  X  +  v
1-Creat Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>|
```



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CREATE

```
D:\CSProject\CRUD.exe
1-Creat Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>1
=====
Student SR Code: 20-00000
Student Name (Last name, First name M.I): espartinez, elaiza v.
Program & Section (BSCS-1101): bscs-1101
Phone Number: 00000008910
Address(Brgy, City/Municipality): Alangilan, Batangas city|

D:\CSProject\CRUD.exe
Current Record/s
=====
No. |SRCODE |STUDENT NAME |Program |Phone Number |Address
-----
1 |20-00000 |ESPARTINEZ, ELAIZA V. |BSCS-1101 |00000008910 |ALANGILAN, BATANGAS CITY
=====
1-Creat Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>1
=====
Student SR Code: 21-00000
Student Name (Last name, First name M.I): Espartinez, elai v.
Program & Section (BSCS-1101): bsit-1101
Phone Number: 12345678901
Address(Brgy, City/Municipality): Kumintang, Batangas City|
```

DISPLAY

```
D:\CSProject\CRUD.exe
Current Record/s
=====
No. |SRCODE |STUDENT NAME |Program |Phone Number |Address
-----
1 |20-00000 |ESPARTINEZ, ELAIZA V. |BSCS-1101 |00000008910 |ALANGILAN, BATANGAS CITY
2 |21-00000 |ESPARTINEZ, ELAI V. |BSIT-1101 |12345678901 |KUMINTANG, BATANGAS CITY
=====
1-Creat Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>|
```

UPDATE

```
1-Creat Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>2|
```

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```
D:\CSProject\CRUD.exe
Current Record/s
=====
No. |SRCODE |STUDENT NAME |Program |Phone Number |Address
=====
1 | 20-00000 | ESPARTINEZ, ELAIZA V. | BSCE-1101 | 000000000010 | ALANGILAN, BATANGAS CITY
2 | 21-00000 | ESPARTINEZ, ELAI V. | BSIT-1101 | 12345678901 | HUMINTANG, BATANGAS CITY
=====
1-Create Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>2
=====
Search by SR Code: 20-00000

D:\CSProject\CRUD.exe
New SR Code: 22-00000
Student Name (Last name, First name, M.I): Elaiza, espartinez v.
New Program: bsce-1101
New Phone Number: 10928374656
New Address: Tinga, Batangas city
-----
Update Successfull!

D:\CSProject\CRUD.exe
Current Record/s
=====
No. |SRCODE |STUDENT NAME |Program |Phone Number |Address
=====
1 | 22-00000 | ELAIZA, ESPARTINEZ V. | BSCE-1101 | 10928374656 | TINGA, BATANGAS CITY
2 | 21-00000 | ESPARTINEZ, ELAI V. | BSIT-1101 | 12345678901 | HUMINTANG, BATANGAS CITY
=====
1-Create Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>|
```



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DELETE:

```
D:\CSProject\CRUD.exe x + v
Current Record/s
=====
No. |SRCODE |STUDENT NAME |Program |Phone Number |Address
=====
1 | 22-00000 |ELAIZA, ESPARTINEZ V. |BSCE-1101 |10928374656 |TINGA, BATANGAS CITY
2 | 21-00000 |ESPARTINEZ, ELAI V. |BSIT-1101 |12345678901 |HUMINTANG, BATANGAS CITY
=====
1-Creat Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>3
=====
Delete by SR Code: |

D:\CSProject\CRUD.exe x + v
Current Record/s
=====
No. |SRCODE |STUDENT NAME |Program |Phone Number |Address
=====
1 | 22-00000 |ELAIZA, ESPARTINEZ V. |BSCE-1101 |10928374656 |TINGA, BATANGAS CITY
2 | 21-00000 |ESPARTINEZ, ELAI V. |BSIT-1101 |12345678901 |HUMINTANG, BATANGAS CITY
=====
1-Creat Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>3
=====
Delete by SR Code: 21-00000
=====
Successfully Deleted!
=====
|
```

```
D:\CSProject\CRUD.exe x + v
Current Record/s
=====
No. |SRCODE |STUDENT NAME |Program |Phone Number |Address
=====
1 | 22-00000 |ELAIZA, ESPARTINEZ V. |BSCE-1101 |10928374656 |TINGA, BATANGAS CITY
=====
1-Creat Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>4
```



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EXIT & SAVE:

```
Current Record/s
=====
No. |SRCODE |STUDENT NAME |Program |Phone Number |Address
=====
1 | 22-00000 | ELAIZA, ESPARTINEZ V. | BSCE-1101 | 10928374656 | TINGA, BATANGAS CITY
=====

1-Creat Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>6
=====
Exit...Saving the Data to File!

Process returned 0 (0x0) execution time : 1975.624 s
Press any key to continue.
```

```
Current Record/s
=====
No. |SRCODE |STUDENT NAME |Program |Phone Number |Address
=====
1 | 22-00000 | ELAIZA, ESPARTINEZ V. | BSCE-1101 | 10928374656 | TINGA, BATANGAS CITY
=====

1-Creat Records
2-Update Records
3-Delete Records
4-Search Records
5-Display All Records
6-Exit and Save to Textfile
=====
Select Option >>6
=====
```

CSProjectSTUDENTFILES.txt

No.	SRCODE	STUDENT NAME	Program	Phone Number	Address
1	22-00000	ELAIZA, ESPARTINEZ V.	BSCE-1101	10928374656	TINGA, BATANGAS CITY

IV. Code

```
#include <iostream>
#include <string>
#include <fstream>
#include <cstdlib>
#include <iomanip>
#include <vector>
```

```
using namespace std;
```

```
const int maxrow = 100;
```

```
string StuName[maxrow] = {};
```

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```
string SRCode[maxrow] = {};  
string Prog[maxrow] = {};  
string Pnum[maxrow] = {};  
string Addr[maxrow] = {};  
  
void OpenFile() {  
    string line;  
    ifstream myfile("D:\\CSProject\\STUDENTFILES.txt");  
  
    if (myfile.is_open()) {  
        int x = 0;  
        while (getline(myfile, line)) {  
            if (line.empty()) continue;  
  
            SRCode[x] = line.substr(0, 18);  
            StuName[x] = line.substr(18, 40);  
            Prog[x] = line.substr(58, 25);  
            Pnum[x] = line.substr(83, 20);  
            Addr[x] = line.substr(103);  
  
            SRCode[x] = SRCode[x].substr(0, SRCode[x].find_last_not_of(' ') + 1);  
            StuName[x] = StuName[x].substr(0, StuName[x].find_last_not_of(' ') + 1);  
            Prog[x] = Prog[x].substr(0, Prog[x].find_last_not_of(' ') + 1);  
            Pnum[x] = Pnum[x].substr(0, Pnum[x].find_last_not_of(' ') + 1);  
            Addr[x] = Addr[x].substr(0, Addr[x].find_last_not_of(' ') + 1);  
  
            x++;  
        }  
  
        myfile.close();  
    } else {  
        cout << "Unable to open file!" << endl;  
    }  
}  
  
void AddRecord(){  
    char name[50];  
    char srcode[10];  
    char prog[50];  
    char pnum[20];  
    char add[50];  
  
    cin.ignore();  
  
    cout << "Student SR Code: ";  
    cin.getline(srcode, 10);  
    cout << "Student Name (Last name, First name M.I): ";
```

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```
cin.getline(name, 50);
cout << "Program & Section (BSCS-1101): ";
cin.getline(prog, 50);
cout << "Phone Number: ";
cin.getline(pnum, 20);
cout << "Address(Brgy, City/Municipality): ";
cin.getline(add, 50);

for (int i = 0; name[i] != '\0'; i++)
{
    name[i] = toupper(name[i]);
}
for (int i = 0; prog[i] != '\0'; i++)
{
    prog[i] = toupper(prog[i]);
}
for (int i = 0; add[i] != '\0'; i++)
{
    add[i] = toupper(add[i]);
}

for (int x = 0; x < maxrow; x++)
{
    if (SRCode[x] == "\0")
    {
        SRCode[x] = srcode;
        StuName[x] = name;
        Prog[x] = prog;
        Pnum[x] = pnum;
        Addr[x] = add;

        break;
    }
}

void ListRecord() {
    system("CLS");
    cout << "Current Record/s" << endl;
    cout <<
    "=====
    =====
    =====" << endl;

    int counter = 0;
    cout << left; // Align all columns to the left
    cout << setw(6) << "No."
        << setw(19) << "|SRCODE"
```

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```
<< setw(41) << "|STUDENT NAME"
<< setw(26) << "|Program"
<< setw(21) << "|Phone Number"
<< "|Address" << endl;
cout << "-----"
-----" << endl;

for (int x = 0; x < maxrow; x++) {
    if (SRCode[x] != "\0") {
        counter++;

        cout << left; // Ensure all data is left-aligned
        cout << setw(8) << counter
            << setw(19) << SRCode[x]
            << setw(41) << StuName[x]
            << setw(26) << Prog[x]
            << setw(21) << Pnum[x]
            << Addr[x] << endl; // Address column does not need setw as it's the last column
    }
}

if (counter == 0) {

    cout << "NO RECORDS FOUND!!" << endl;
}

cout <<
"=====
=====
===== " << endl;
}

void SearchRecord(string search) {
    system("CLS");
    cout << "Current Record/s" << endl;
    cout <<
"=====
=====
===== " << endl;

    int counter = 0;
    cout << left; // Align all columns to the left
    cout << setw(6) << "No. "
        << setw(19) << "|SRCODE "
        << setw(41) << "|STUDENT NAME "
        << setw(26) << "|Program "
        << setw(21) << "|Phone Number "
        << "|Address" << endl;
```

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```
cout << "-----" << endl;

for (int x = 0; x < maxrow; x++) {
    if (SRCode[x] == search) {
        counter++;
        cout << left; // Ensure all data is left-aligned
        cout << setw(8) << counter
            << setw(19) << SRCode[x]
            << setw(41) << StuName[x]
            << setw(26) << Prog[x]
            << setw(21) << Pnum[x]
            << Addr[x] << endl;
        break; // Exit loop after finding the record
    }
}

if (counter == 0) {
    cout << "NO RECORD FOUND!" << endl;
}

cout <<
"=====
=====
===== " << endl;
}

void UpdateRecord(string search){
    system("CLS");
    char name[50];
    char srcode[10];
    char prog[50];
    char pnum[20];
    char add[50];

    int counter = 0;

    for (int x = 0; x < maxrow; x++)
    {
        if (SRCode[x] == search)
        {
            counter++;

            cout << "New SR Code: ";
            cin.getline(srcode, 10);

            SRCode[x] = srcode;
```

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```
cout << "Student Name (Last name, First name, M.I): ";
cin.getline(name, 50);

for (int i = 0; name[i] != '\0'; i++)
{
    name[i] = toupper(name[i]);
}

StuName[x] = name;

cout << "New Program: ";
cin.getline(prog, 50);

for (int i = 0; prog[i] != '\0'; i++)
{
    prog[i] = toupper(prog[i]);
}

Prog[x] = prog;

cout << "New Phone Number: ";
cin.getline(pnum, 20);

Pnum[x] = pnum;

cout << "New Address: ";
cin.getline(add, 50);

for (int i = 0; add[i] != '\0'; i++)
{
    add[i] = toupper(add[i]);
}

Addr[x] = add;

cout << "-----" << endl;
cout << "Update Successfull!" << endl;
cout << "===== " << endl;
break;
}
}

if (counter == 0){
    cout << "SRCODE DOES NOT EXIST!" << endl;
    cout << "===== " << endl;
```

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```
}  
}  
  
void DeleteRecord(string search) {  
    int counter = 0;  
    for (int x = 0; x < maxrow; x++) {  
        if (SRCode[x] == search) {  
            counter++;  
  
            for (int y = x; y < maxrow - 1; y++) {  
                SRCode[y] = SRCode[y + 1];  
                StuName[y] = StuName[y + 1];  
                Prog[y] = Prog[y + 1];  
                Pnum[y] = Pnum[y + 1];  
                Addr[y] = Addr[y + 1];  
            }  
  
            SRCode[maxrow - 1] = "";  
            StuName[maxrow - 1] = "";  
            Prog[maxrow - 1] = "";  
            Pnum[maxrow - 1] = "";  
            Addr[maxrow - 1] = "";  
  
            cout << "-----" << endl;  
            cout << "Successfully Deleted!" << endl;  
            cout << "=====" << endl;  
            break;  
        }  
    }  
    if (counter == 0) {  
        cout << "=====" << endl;  
        cout << "SR CODE DOES NOT EXIST!" << endl;  
        cout << "=====" << endl;  
    }  
}  
  
void SaveData() {  
    ofstream myfile;  
    myfile.open("D:\\CSProject\\STUDENTFILES.txt");  
  
    if (!myfile.is_open()) {  
        cout << "Error: Unable to save data to file!" << endl;  
        return;  
    }  
  
    for (int x = 0; x < maxrow; x++) {  
        if (SRCode[x] == "\\0") {  
            break;  
        }  
    }  
}
```

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```
} else {
    // Write data in a properly aligned format
    myfile << left
        << setw(18) << SRCode[x]
        << setw(40) << StuName[x]
        << setw(25) << Prog[x]
        << setw(20) << Pnum[x]
        << Addr[x] << endl; // Address column doesn't need setw as it's the last
    }
}

myfile.close();
}

int main (){

    cout << "MENU\n";
    int option;
    string srcode;
    system("CLS");
    OpenFile();

    do {
        cout << "1-Creat Record" << endl;
        cout << "2-Update Record" << endl;
        cout << "3-Delete Record" << endl;
        cout << "4-Search Record" << endl;
        cout << "5-Display All Record" << endl;
        cout << "6-Exit and Save to Textfile" << endl;
        cout << "===== " << endl;

        cout << "Select Option >>";
        cin >> option;
        cout << "===== " << endl;

        switch (option)
        {
            case 1:
                AddRecord();
                system("CLS");
                break;

            case 2:
                cin.ignore();
                cout << "Search by SR Code: ";
                getline(cin, srcode);
                UpdateRecord(srcode);
                break;
```

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case 3:

```
cin.ignore();  
cout << "Delete by SR Code: ";  
getline(cin, srcode);  
DeleteRecord(srcode);  
cin.ignore();  
system("CLS");  
break;
```

case 4:

```
cin.ignore();  
cout << "Search by SR Code: ";  
getline(cin, srcode);  
SearchRecord(srcode);  
break;
```

case 5:

```
ListRecord();  
break;
```

}

} while (option !=6);

```
SaveData();  
cout << "Exit....Saving the Data to File!" << endl;  
return 0;  
}
```